

```
>> rundcpf('case14')
```

MATPOWER Version 8.1, 12-Jul-2025

Power Flow -- DC formulation

PF successful

Converged in 0.28 seconds

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|      System Summary      |
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How many?		How much?	P (MW)	Q (MVar)
Buses	14	Total Gen Capacity	772.4	0.0 to 0.0
Generators	5	On-line Capacity	772.4	0.0 to 0.0
Committed Gens	5	Generation (actual)	259.0	0.0
Loads	11	Load	259.0	0.0
Fixed	11	Fixed	259.0	0.0
Dispatchable	0	Dispatchable	-0.0 of -0.0	-0.0
Shunts	0	Shunt (inj)	-0.0	0.0
Branches	20	Losses ($I^2 * Z$)	0.00	0.00
Transformers	3	Branch Charging (inj)	-	0.0
Inter-ties	0	Total Inter-tie Flow	0.0	0.0
Areas	1			

	Minimum	Maximum
Voltage Magnitude	1.000 p.u. @ bus 1	1.000 p.u. @ bus 1
Voltage Angle	-17.19 deg @ bus 14	0.00 deg @ bus 1

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|      Bus Data      |
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Bus #	Voltage Mag(pu)	Voltage Ang(deg)	Generation P (MW)	Generation Q (MVar)	Load P (MW)	Load Q (MVar)
1	1.000	0.000*	219.00	0.00	-	-
2	1.000	-5.012	40.00	0.00	21.70	0.00
3	1.000	-12.954	0.00	0.00	94.20	0.00
4	1.000	-10.584	-	-	47.80	0.00
5	1.000	-9.094	-	-	7.60	0.00
6	1.000	-14.852	0.00	0.00	11.20	0.00
7	1.000	-13.907	-	-	-	-
8	1.000	-13.907	0.00	0.00	-	-
9	1.000	-15.695	-	-	29.50	0.00
10	1.000	-15.974	-	-	9.00	0.00
11	1.000	-15.619	-	-	3.50	0.00
12	1.000	-15.967	-	-	6.10	0.00
13	1.000	-16.140	-	-	13.50	0.00
14	1.000	-17.188	-	-	14.90	0.00
Total:			259.00	0.00	259.00	0.00

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|      Branch Data      |
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Brnch #	From Bus	To Bus	From Bus Injection		To Bus Injection		Loss ($I^2 * Z$)	
			P (MW)	Q (MVar)	P (MW)	Q (MVar)	P (MW)	Q (MVar)
1	1	2	147.84	0.00	-147.84	0.00	0.000	0.00
2	1	5	71.16	0.00	-71.16	0.00	0.000	0.00
3	2	3	70.01	0.00	-70.01	0.00	0.000	0.00
4	2	4	55.15	0.00	-55.15	0.00	0.000	0.00
5	2	5	40.97	0.00	-40.97	0.00	0.000	0.00
6	3	4	-24.19	0.00	24.19	0.00	0.000	0.00
7	4	5	-61.75	0.00	61.75	0.00	0.000	0.00
8	4	7	28.36	0.00	-28.36	0.00	0.000	0.00
9	4	9	16.55	0.00	-16.55	0.00	0.000	0.00
10	5	6	42.79	0.00	-42.79	0.00	0.000	0.00
11	6	11	6.73	0.00	-6.73	0.00	0.000	0.00
12	6	12	7.61	0.00	-7.61	0.00	0.000	0.00
13	6	13	17.25	0.00	-17.25	0.00	0.000	0.00
14	7	8	0.00	0.00	0.00	0.00	0.000	0.00
15	7	9	28.36	0.00	-28.36	0.00	0.000	0.00
16	9	10	5.77	0.00	-5.77	0.00	0.000	0.00
17	9	14	9.64	0.00	-9.64	0.00	0.000	0.00
18	10	11	-3.23	0.00	3.23	0.00	0.000	0.00
19	12	13	1.51	0.00	-1.51	0.00	0.000	0.00
20	13	14	5.26	0.00	-5.26	0.00	0.000	0.00

Total: 0.000 0.00

```
>> mpc = case14; mpc.gen(2,2) = 46.67;mpc.gen(3,2) = 4.83;mpc.gen(4,2) = 1.79;mpc.gen(5,2) = 0;
>> results = runpf(mpc);
```

MATPOWER Version 8.1, 12-Jul-2025
Power Flow -- AC-polar-power formulation

Newton's method converged in 2 iterations.
PF successful

Converged in 0.20 seconds

System Summary

How many?		How much?		P (MW)	Q (MVar)
Buses		Total Gen Capacity		772.4	-52.0 to 148.0
Generators		On-line Capacity		772.4	-52.0 to 148.0
Committed Gens		Generation (actual)		271.2	78.1
Loads		Load		259.0	73.5
Fixed		Fixed		259.0	73.5
Dispatchable		Dispatchable		-0.0 of -0.0	-0.0
Shunts		Shunt (inj)		-0.0	21.2
Branches		Losses ($I^2 * Z$)		12.23	50.24
Transformers		Branch Charging (inj)		-	24.4
Inter-ties		Total Inter-tie Flow		0.0	0.0
Areas					

Minimum

Maximum

Voltage Magnitude 1.010 p.u. @ bus 3 1.090 p.u. @ bus 8

Voltage Angle -15.51 deg @ bus 14 0.00 deg @ bus 1
P Losses ($I^2 \cdot R$) - 3.70 MW @ line 1-2
Q Losses ($I^2 \cdot X$) - 11.29 MVar @ line 1-2

Bus Data						
Bus #	Voltage		Generation		Load	
	Mag(pu)	Ang(deg)	P (MW)	Q (MVar)	P (MW)	Q (MVar)
1	1.060	0.000*	217.94	-13.80	-	-
2	1.045	-4.612	46.67	39.19	21.70	12.70
3	1.010	-12.034	4.83	22.93	94.20	19.00
4	1.018	-9.867	-	-	47.80	-3.90
5	1.020	-8.380	-	-	7.60	1.60
6	1.070	-13.656	1.79	12.24	11.20	7.50
7	1.062	-12.884	-	-	-	-
8	1.090	-12.884	0.00	17.58	-	-
9	1.056	-14.447	-	-	29.50	16.60
10	1.051	-14.593	-	-	9.00	5.80
11	1.057	-14.256	-	-	3.50	1.80
12	1.055	-14.515	-	-	6.10	1.60
13	1.050	-14.601	-	-	13.50	5.80
14	1.036	-15.515	-	-	14.90	5.00
Total:			271.23	78.14	259.00	73.50

Branch Data								
Brnch #	From Bus	To Bus	From Bus P (MW)	Injection Q (MVar)	To Bus P (MW)	Injection Q (MVar)	Loss (I^2 * Z)	
							P (MW)	Q (MVar)
1	1	2	145.68	-17.75	-141.98	23.19	3.698	11.29
2	1	5	72.27	3.95	-69.73	1.19	2.533	10.46
3	2	3	70.36	3.85	-68.21	0.57	2.147	9.04
4	2	4	55.36	-1.55	-53.73	2.88	1.631	4.95
5	2	5	41.23	1.00	-40.34	-1.97	0.891	2.72
6	3	4	-21.16	3.36	21.47	-3.90	0.305	0.78
7	4	5	-59.27	14.88	59.75	-13.36	0.481	1.52
8	4	7	27.81	-9.58	-27.81	11.25	0.000	1.67
9	4	9	15.93	-0.38	-15.93	1.66	0.000	1.28
10	5	6	42.72	12.54	-42.72	-8.37	0.000	4.17
11	6	11	7.61	3.46	-7.56	-3.34	0.058	0.12
12	6	12	7.82	2.49	-7.74	-2.33	0.072	0.15
13	6	13	17.88	7.17	-17.67	-6.75	0.214	0.42
14	7	8	-0.00	-17.12	0.00	17.58	0.000	0.46
15	7	9	27.81	5.87	-27.81	-5.08	-0.000	0.79
16	9	10	4.97	4.33	-4.96	-4.29	0.012	0.03
17	9	14	9.26	3.68	-9.15	-3.44	0.113	0.24
18	10	11	-4.04	-1.51	4.06	1.54	0.014	0.03
19	12	13	1.64	0.73	-1.64	-0.73	0.006	0.01
20	13	14	5.81	1.67	-5.75	-1.56	0.057	0.12
Total:							12.233	50.24

>> runopf('case14')

OPF successful

Converged in 1.60 seconds

Objective Function Value = 8081.53 \$/hr

System Summary

How many?		How much?	P (MW)	Q (MVar)
Buses	14	Total Gen Capacity	772.4	-52.0 to 148.0
Generators	5	On-line Capacity	772.4	-52.0 to 148.0
Committed Gens	5	Generation (actual)	268.3	67.6
Loads	11	Load	259.0	73.5
Fixed	11	Fixed	259.0	73.5
Dispatchable	0	Dispatchable	-0.0 of -0.0	-0.0
Shunts	1	Shunt (inj)	-0.0	20.7
Branches	20	Losses ($I^2 * Z$)	9.29	39.16
Transformers	3	Branch Charging (inj)	-	24.3
Inter-ties	0	Total Inter-tie Flow	0.0	0.0
Areas	1			

	Minimum	Maximum
Voltage Magnitude	1.014 p.u. @ bus 4	1.060 p.u. @ bus 1
Voltage Angle	-14.27 deg @ bus 14	0.00 deg @ bus 1
P Losses (I^2R)	-	2.90 MW @ line 1-2
Q Losses (I^2X)	-	8.86 MVar @ line 1-2
Lambda P	36.72 \$/MWh @ bus 1	41.20 \$/MWh @ bus 14
Lambda Q	-0.09 \$/MVarh @ bus 1	0.57 \$/MVarh @ bus 14

Bus Data

Bus #	Voltage		Generation		Load		Lambda(\$/MVA-hr)	
	Mag(pu)	Ang(deg)	P (MW)	Q (MVar)	P (MW)	Q (MVar)	P	Q
1	1.060	0.000*	194.33	0.00	-	-	36.724	-0.094
2	1.041	-4.022	36.72	23.69	21.70	12.70	38.360	-
3	1.016	-9.926	28.74	24.13	94.20	19.00	40.575	-
4	1.014	-8.665	-	-	47.80	-3.90	40.190	0.120
5	1.016	-7.428	-	-	7.60	1.60	39.661	0.208
6	1.060	-12.689	0.00	11.55	11.20	7.50	39.734	-
7	1.046	-11.188	-	-	-	-	40.172	0.120
8	1.060	-10.415	8.49	8.27	-	-	40.170	-
9	1.044	-12.997	-	-	29.50	16.60	40.166	0.196
10	1.039	-13.233	-	-	9.00	5.80	40.318	0.309
11	1.046	-13.091	-	-	3.50	1.80	40.155	0.228
12	1.045	-13.533	-	-	6.10	1.60	40.379	0.212
13	1.040	-13.583	-	-	13.50	5.80	40.575	0.353

14 1.024 -14.274 - - 14.90 5.00 41.197 0.571

Total: 268.29 67.63 259.00 73.50

Branch Data

Brnch #	From Bus	To Bus	From Bus P (MW)	Injection Q (MVA _r)	To Bus P (MW)	Injection Q (MVA _r)	Loss (I ² * Z)	
							P (MW)	Q (MVA _r)
1	1	2	129.67	-6.36	-126.77	9.40	2.902	8.86
2	1	5	64.66	6.37	-62.61	-3.21	2.051	8.47
3	2	3	55.59	0.47	-54.25	0.56	1.344	5.66
4	2	4	48.92	-0.48	-47.63	0.79	1.285	3.90
5	2	5	37.28	1.59	-36.54	-3.00	0.737	2.25
6	3	4	-11.21	4.57	11.31	-5.63	0.099	0.25
7	4	5	-49.17	11.58	49.50	-10.53	0.331	1.04
8	4	7	22.85	-3.99	-22.85	5.04	0.000	1.05
9	4	9	14.84	1.17	-14.84	-0.04	0.000	1.12
10	5	6	42.06	15.14	-42.06	-10.90	0.000	4.23
11	6	11	6.09	4.56	-6.04	-4.46	0.049	0.10
12	6	12	7.65	2.66	-7.58	-2.51	0.072	0.15
13	6	13	17.12	7.73	-16.91	-7.32	0.208	0.41
14	7	8	-8.49	-8.05	8.49	8.27	0.000	0.22
15	7	9	31.34	3.01	-31.34	-2.02	-0.000	1.00
16	9	10	6.49	3.20	-6.47	-3.16	0.015	0.04
17	9	14	10.20	2.95	-10.06	-2.67	0.131	0.28
18	10	11	-2.53	-2.64	2.54	2.66	0.010	0.02
19	12	13	1.48	0.91	-1.47	-0.91	0.006	0.01
20	13	14	4.88	2.42	-4.84	-2.33	0.047	0.10
Total:							9.287	39.16

Voltage Constraints

Bus #	Vmin mu	Vmin	V	Vmax	Vmax mu
1	-	0.940	1.060	1.060	583.780
6	-	0.940	1.060	1.060	55.184
8	-	0.940	1.060	1.060	71.069

Generation Constraints

Gen #	Bus #	Pmin mu	Pmin	Pg	Pmax	Pmax mu
4	6	0.266	0.00	0.00	100.00	-

Gen #	Bus #	Qmin mu	Qmin	Qg	Qmax	Qmax mu
1	1	0.094	0.00	0.00	10.00	-

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